Approximating Barnette's Conjecture

Michael Kaufmann
with Michalis Bekos, Maximilian Pfister

GD 2025

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Hamiltonicity in planar graphs

Long history ended in

Barnette's conjecture:
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- true for graphs generated only by restricted operations

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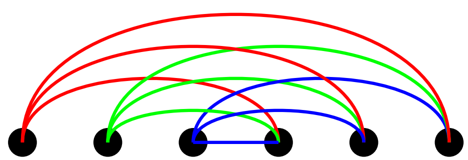
- true for graphs generated only by restricted operations

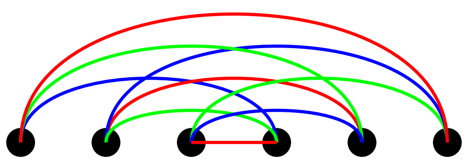
Our result:
Every Barnette graph has a subhamiltonean cycle
with at least $\frac{5}{6}n$ edges.

Our previous work

- Jawaherul Alam, Michael A. Bekos, Vida Dujmovic, Martin Gronemann, MK, Sergey Pupyrev: On dispersable book embeddings. Theor. Comput. Sci. 861: 1-22 (2021)

Dispersable:





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Find another order !

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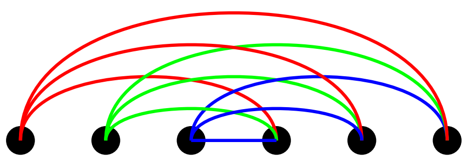
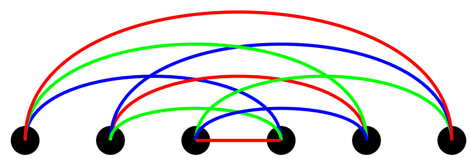


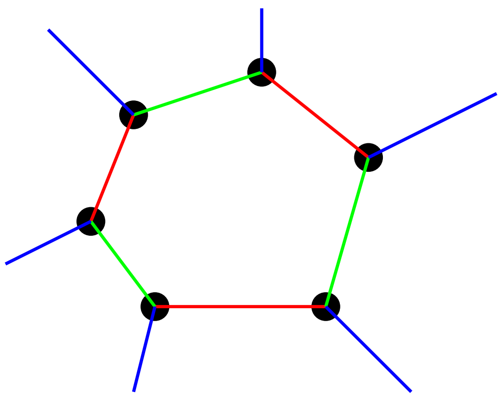


Find another order !

Any 3-connected cubic bipartite planar graph is dispersable !

From book embedding to subhamiltonicity

Algorithm constructs a 3-edge coloring s.t.
faces have only 2 colors at their boundary



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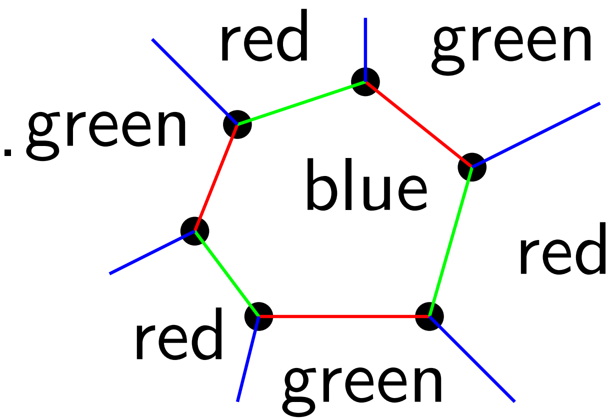

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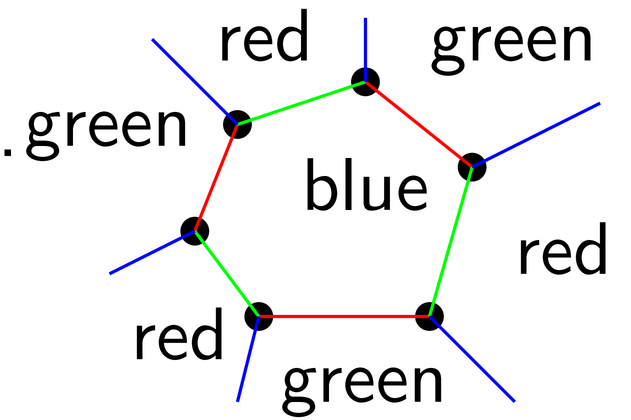

Build a spanning tree T_{bg} in the dual crossing only red edges.

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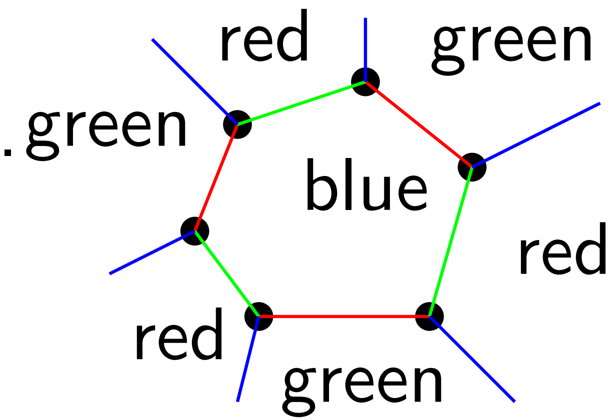

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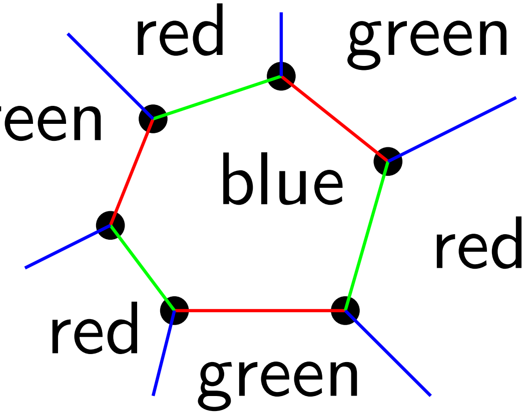

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Build a spanning tree T_{bg} in the dual crossing only red edges.

Walk along T_{bg} and construct a subhamiltonean cycle.

– processing blue faces, we take all edges (green, uncrossed red)
– in green faces, we take blue edges only if between two crossed red edges or if green leaf..

Properties of cycle C

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Hence in total $n/2 + n/6 + x > \frac{2}{3}n$ edges.

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If leaf, then a red (crossed) edge, if no-leaf, then a blue edge.

Choose colors s.t. $|F_g| \geq |F_r| \geq |F_b|$.
$\Rightarrow |F_g| = \frac{n}{6} + x$, $|F_r| \geq \frac{n}{6} - \frac{x}{2}$

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Hence, C contains $\geq \frac{n}{2} + \frac{n}{6} - \frac{x}{2} + \frac{n}{6} + x = \frac{5}{6}n + \frac{x}{2}$ edges

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Thank you !